1) CLASS LAYOUTS

Namespace numcomp

BasicRootFinder Owner: Jairus
(no data member: stateless
+ bisection (f, a, b, tol, max_iter + Newton_raphson (f, df, x0, tol, max_iter)

AdvancedRootFinder Owner: Valeria
(no data member: stateless)
+ secant_method (f, x0, x1, tol, max_iter) + fixed_point_interation (g, x0, tol, max_iter)

NumericalDifferentiator Owner: Jairus (week5)
(no data member: stateless
+ forward_difference (f, x, h) + backward_difference (f, x, h) + central_difference (f, x, h) + second_derivative (f, x, h) + richardson_extrapolation (f, x, h)

interpolator Owner: Derrick
(no data member: stateless
+ linear_interpolator (x, x_vals, y_vals) + chebyshev_nodes (a, b, n) + chebyshev_coefficients (f, a, b, n) + chebyshev_approximate (coeffs, a, b, x)

cubicspline Owner: Derrick (week4)
(no data member: stateless)
+ second_derivatives (x_vals, y_vals) + interpolate (x, x_vals, y_vals

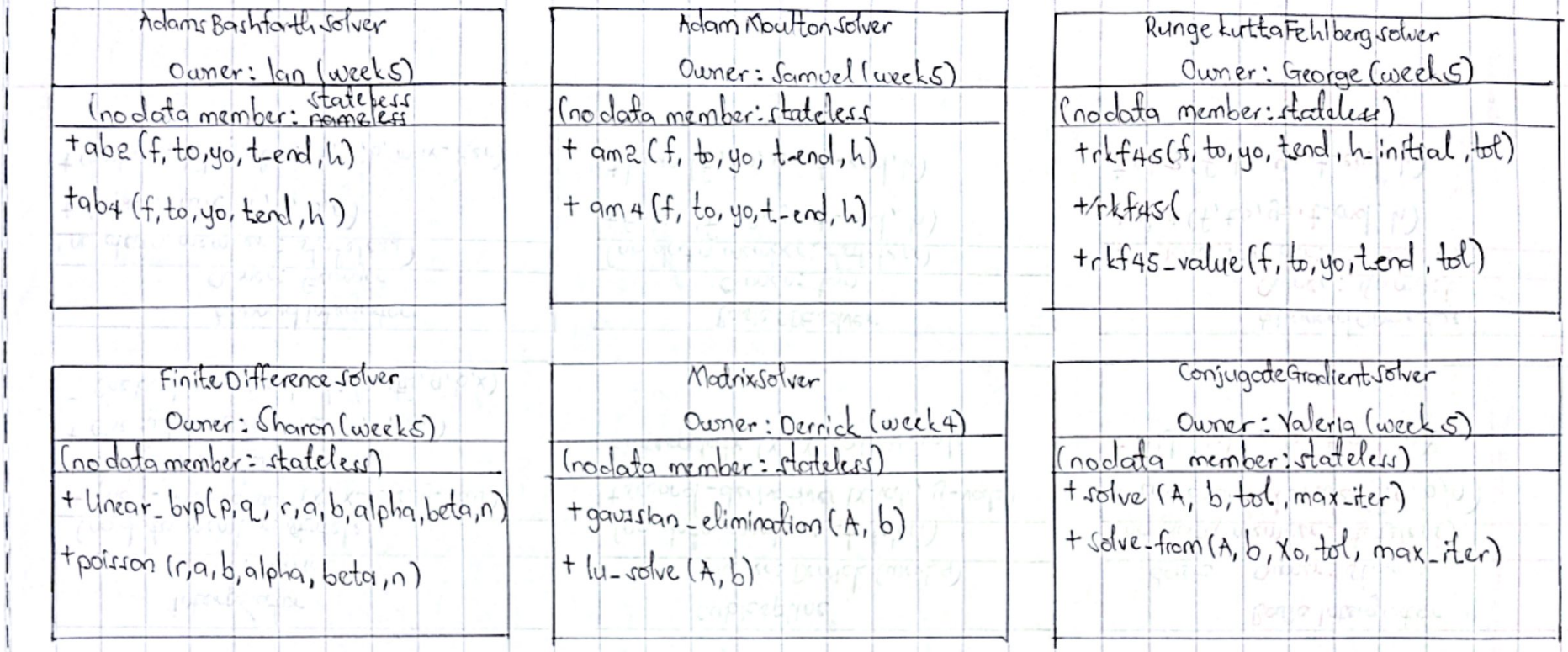
Basic Intergrator Basic Owner: sharon
(no data member: stateless)
+ trapezoidal_rule (f, a, b, n) + midpoint_rule (f, a, b, n)

Advanced Integrator Owner: Georger
(no data member: stateless)
+ rimpsons_rule (f, a, b, n) + romberg_integration (f, a, b, max_iter)

Basic ODEsolver Owner: Ian
(no data member: stateless)
+ Euler (f, t0, y0, t_end, h) + heun (f, t0, y0, t_end, h)

Advanced ODEsolver Owner: Samuel
(no data member: stateless)
+ rk4 (f, t0, y0, t_end, h) + rk3 (f, t0, y0, t_end, h)

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